

1. Write a C program to implement Dijkstra's algorithm for the single source shortest path problem.

```
int cost[10][10];
void dijkstra(int,int);
void main()
{
    int i,j,n,u;
    printf("Enter no. of vertices:");
    scanf("%d",&n);
    printf("\nEnter the cost matrix:\nEnter 9999 if there is no edge between the
vertices\n");
    for(i=0;i<n;i++)
        for(j=0;j<n;j++)
            scanf("%d",&cost[i][j]);
    printf("\nEnter the starting node:");
    scanf("%d",&u);
    dijkstra(n,u);
}
void dijkstra(int n,int startnode)
{
    int distance[10];
    int visited[10],mindistance,nextnode,i,j,count;
    for(i=0;i<n;i++)
    {
        distance[i]=cost[startnode][i];
        visited[i]=0;
    }
    distance[startnode]=0;
    visited[startnode]=1;
    count=1;
    while(count<n-1)
    {
        mindistance=9999; //INFINITY 9999
        for(i=0;i<n;i++)
            if(distance[i]<mindistance&&!visited[i])
            {
                mindistance=distance[i];
                nextnode=i;
            }
        visited[nextnode]=1;
        for(i=0;i<n;i++)
            if(!visited[i])
```

```

        if(mindistance+cost[nextnode][i]<distance[i])
        {
            distance[i]=mindistance+cost[nextnode][i];
        }
        count++;
    }
    for(i=0;i<n;i++)
        if(i!=startnode)
            printf("\nDistance from node%d to node%d=%d",startnode,i,distance[i]);
}

```

Output:

Enter no. of vertices: 6

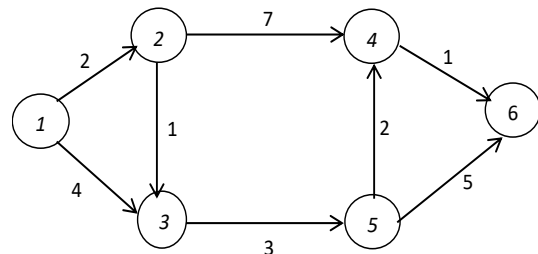
Enter the cost matrix:

Enter 9999 if there is no edge between the vertices

```

9999 2 4 9999 9999 9999
9999 9999 1 7 9999 9999
9999 9999 9999 9999 3 9999
9999 9999 9999 9999 9999 1
9999 9999 9999 2 9999 5
9999 9999 9999 9999 9999 9999

```



Enter the starting node: 0

Distance from node0 to node1=2

Distance from node0 to node2=3

Distance from node0 to node3=8

Distance from node0 to node4=6

Distance from node0 to node5=9